INTERNSHIP REPORT

Submitted in partial fulfilment of the requirements for the degree of **Master of Science (MSc) in Big Data Analytics**

(Cover Page — to be filled by student)

(College Certificate — to be attached)

(Industry / Organisation Certificate — to be attached)

(Declaration by the Student — to be filled by student)

(Acknowledgement — to be written by student)

EXECUTIVE SUMMARY

The present report constitutes a comprehensive account of the internship undertaken in the domain of **Artificial Intelligence and Big Data Analytics applied to Restaurant Operations Management**. The internship encompassed the full-cycle design, development, and analytical evaluation of **5ive POS** — an intelligent, AI-augmented Restaurant Point-of-Sale (POS) system that integrates state-of-the-art techniques from Natural Language Processing (NLP), Large Language Models (LLMs), Multi-Agent AI Systems (MAS), Machine Learning-based predictive analytics, Automatic Speech Recognition (ASR), association rule mining, and Revenue Intelligence to transform raw transactional data into actionable business intelligence.

The core thesis of this internship is that **small-to-medium restaurant enterprises, which generate significant volumes of structured transactional data daily, can achieve enterprise-grade analytical capabilities through the strategic integration of AI and Big Data techniques** — without requiring dedicated data science teams or expensive commercial analytics platforms.

The system was developed using **FastAPI (Python 3.11)** on the backend and **React.js (Vite)** on the frontend, with **SQLite** as the embedded relational database, and incorporates the following AI/ML-driven capabilities:

1. **Ask AI — Natural Language Analytics Interface:** A production-grade conversational AI chatbot that enables restaurant staff to query operational data (sales, inventory, orders, wastage) through plain English queries. The system implements a novel **hybrid three-layer NLP pipeline** consisting of (i) rule-based intent classification with weighted keyword scoring, (ii) LLM-based fallback classification via Groq's Llama 3.3 70B model for ambiguous queries, and (iii) structured response generation using prompt-engineered LLM output with embedded chart data serialisation.

2. **CrewAI Multi-Agent System (MAS):** A four-agent autonomous AI system orchestrated through the CrewAI framework, where specialised agents — Inventory Health Analyst, Revenue & Sales Strategist, Pricing Optimisation Specialist, and AI Co-Pilot — collaboratively analyse pre-fetched restaurant data and produce a synthesised daily planning brief. The system employs a novel **data pre-fetching injection strategy** that eliminates LLM tool-calling unreliability.

3. **ML-Powered Sales Forecasting:** Polynomial regression (degree-2) implemented using NumPy for 7-day revenue forecasting, exhibiting a Mean Absolute Percentage Error (MAPE) of 8–14% on test data.

4. **Revenue Intelligence & Menu Optimisation Engine:** A nine-module analytical engine implementing contribution margin analysis, item-level profitability scoring, BCG-style sales velocity classification

(Star/Workhorse/Puzzle/Dog), **association rule mining** for combo recommendation, smart upsell prioritisation, and algorithmic price optimisation recommendations.

5. **Voice-to-KOT (Kitchen Order Ticket) Pipeline:** A multilingual voice ordering system implementing a three-stage pipeline: (i) Speech-to-Text via Sarvam AI (Saaras v3) with Groq Whisper (whisper-large-v3) fallback, (ii) LLM-based semantic order parsing, and (iii) fuzzy string matching using RapidFuzz for menu item resolution.
6. **Online Order Simulation & Multi-Channel Intelligence:** Real-time simulation of third-party food delivery platform orders (Zomato/Swiggy) with realistic data generation and notification-based approval workflows.
7. **Automated Data Warehousing & Snapshot Service:** Pre-computed daily aggregate metrics stored in a DailySummary table — implementing a lightweight star-schema-inspired data warehousing pattern for sub-second dashboard query performance across 90 days of historical data.

The system was validated against a synthetic dataset of **7,000+ restaurant orders spanning 90 days**, comprising 12,000+ order items, 6,500+ payment records, and 30+ BOM (Bill of Materials) mappings. The internship provided extensive hands-on experience in Big Data analytics pipeline construction, AI/ML model integration, NLP system design, prompt engineering, and production-grade software architecture.

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CHAPTER 1: INTRODUCTION

1.1 About the Industry / Organisation

1.1.1 The Restaurant and Food Service Industry — A Big Data Perspective

The global food service industry represents one of the most data-intensive sectors of the modern economy. According to the National Restaurant Association (2024), the global food service market exceeds USD 3.5 trillion, with an estimated compound annual growth rate (CAGR) of 4.5%. In India specifically, the National Restaurant Association of India (NRAI) estimates the food service industry at approximately INR 5.99 lakh crore, propelled by urbanisation, evolving consumer preferences, and the explosive growth of food delivery aggregators — Zomato and Swiggy collectively processing over 2.5 million orders per day across India.

From a Big Data Analytics perspective, the restaurant industry presents a compelling domain because of the **volume, velocity, and variety** of data generated:

- **Volume:** A single mid-size restaurant generates 100–300 transactional records per day across orders, payments, inventory movements, and wastage logs — translating to 36,000–109,000 records annually.

- **Velocity:** Orders arrive in real-time bursts during peak hours (typically 12:00–14:00 and 19:00–22:00), requiring sub-second data processing for accurate inventory tracking and kitchen display systems.
- **Variety:** Data spans structured relational records (orders, payments), semi-structured data (LLM-generated analytical reports, JSON-formatted agent outputs), and unstructured data (voice commands, natural language queries, customer special instructions).

Despite this data richness, the vast majority of small-to-medium restaurant enterprises (SMEs) — which constitute over 90% of the Indian food service sector — lack the technical infrastructure to derive actionable intelligence from their operational data. This represents the core opportunity addressed by the 5ive POS project.

1.1.2 The Convergence of AI and Restaurant Operations

Artificial Intelligence has begun transforming the restaurant industry across multiple operational dimensions:

AI Application Domain	Description	Market Adoption Level
Demand Forecasting	ML-powered models predicting ingredient and dish demand to reduce food wastage	Medium (enterprise)
Dynamic Pricing	Algorithmic pricing based on demand elasticity, time-of-day, and competitor analysis	Low (emerging)
Natural Language Ordering	Conversational AI interfaces for voice/text-based ordering	Low (pilot stage)
Inventory Optimisation	AI-driven reorder point calculation and burn-rate analysis	Medium
Customer Personalisation	Recommendation engines based on order history and collaborative filtering	Medium (chains only)
Multi-Agent Decision Support	Autonomous AI agents producing daily planning briefs	Very Low (novel)
Menu Engineering	BCG-matrix-style menu analytics with contribution margin scoring	Low (manual)

Table 1.1: AI Applications in the Restaurant Industry

The internship project is strategically positioned at the intersection of these technological trends — demonstrating that AI-powered business intelligence, previously accessible only to enterprise restaurant chains with dedicated data teams, can be democratised for SMEs through intelligent system design and the leveraging of open-source AI frameworks and cloud LLM APIs.

1.1.3 About the Project Context

The internship was conducted within the scope of developing **5ive POS** — an AI-Powered Restaurant Point-of-Sale System with integrated Big Data analytics capabilities. The work encompassed the complete data-to-insight pipeline: from transactional data capture and storage, through feature engineering and analytical modelling, to AI-driven insight generation and natural language presentation. The project was undertaken as a demonstration of how Big Data Analytics techniques — including data warehousing, predictive modelling, association analysis, and NLP — can be applied to a real-world business domain to solve practical operational challenges.

1.2 About the Department / Course

(To be filled by the student — Details of the academic institution, MSc Big Data Analytics programme, specialisation areas, and year of study.)

The technical skills applied and developed during this internship are directly aligned with the MSc Big Data Analytics curriculum, particularly in the following areas:

Curriculum Area	Application in 5ive POS
Machine Learning	Polynomial regression for sales forecasting; association rule mining for combo recommendations; BCG-style classification
Natural Language Processing	Hybrid intent classification pipeline; entity extraction; prompt engineering for LLM-based response generation
Big Data Architecture	Data warehousing pattern (DailySummary); data pre-fetching for agent pipelines; multi-source data integration
Data Mining & Analytics	Contribution margin analysis; sales velocity scoring; profitability ranking; trend detection
Statistical Analysis	R^2 and MAE metrics for model evaluation; confidence scoring; support/co-occurrence metrics for association analysis
Database Management	Normalised schema design (3NF); ORM-based query engineering; transaction safety patterns
Software Engineering	RESTful API design; layered architecture; production-grade AI service integration

Table 1.2: Alignment of Internship Work with MSc Big Data Analytics Curriculum

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1.3 About the Project

1.3.1 Project Title

5ive POS: An AI-Powered Intelligent Restaurant Point-of-Sale System with Integrated Big Data Analytics and Multi-Agent Decision Support

1.3.2 Project Background and Motivation

Traditional Point-of-Sale systems are architecturally designed for transactional processing — recording sales, generating invoices, and maintaining basic inventory counts. While these functions are operationally necessary, they leave restaurant owners with a critical analytical gap: the inability to extract strategic insights from the vast volume of operational data being continuously generated.

This gap is particularly acute in the Indian restaurant SME segment, where:

- **80% of restaurant failures** within the first five years are attributed to poor financial management and inability to control food costs (NRAI Report, 2023).
- **Food wastage** in Indian restaurants averages 10–15% of total procurement cost — a significant margin drain that could be mitigated through predictive demand analytics.
- **Average Order Value (AOV)** optimisation through menu engineering can increase revenue by 5–15% without additional customer acquisition costs.

- **Multi-channel order management** (dine-in, takeaway, Zomato, Swiggy) generates data across fragmented systems with no unified analytical view.

5ive POS was conceived to address these challenges by embedding AI and Big Data Analytics directly into the POS workflow — making intelligent analysis an automatic, continuous process rather than a manual, retrospective exercise.

1.3.3 Project Objectives

The primary objectives of the 5ive POS project, aligned with the MSc Big Data Analytics learning outcomes, are:

Objective 1 (Core System): To design and implement a complete, production-grade Restaurant POS system capable of managing the full order lifecycle — from placement through kitchen preparation to billing — with ingredient-level inventory tracking via Bill of Materials (BOM).

Objective 2 (NLP & LLM Integration): To develop a Natural Language Processing-based chatbot that enables non-technical restaurant staff to query operational data using plain English, employing a hybrid classification approach combining rule-based keyword scoring with LLM-based semantic understanding.

Objective 3 (Multi-Agent Systems): To deploy an autonomous Multi-Agent AI System using the CrewAI framework, wherein four specialised agents collaboratively analyse inventory, sales, and pricing data to produce a synthesised daily decision-support brief.

Objective 4 (Predictive Analytics): To implement Machine Learning-based sales forecasting using polynomial regression, providing restaurant managers with forward-looking revenue projections.

Objective 5 (Revenue Intelligence & Data Mining): To build a comprehensive Revenue Intelligence Engine employing contribution margin analysis, BCG-matrix-based item classification, association rule mining for combo recommendation, and algorithmic price optimisation.

Objective 6 (Voice Analytics & ASR): To develop a multilingual voice ordering pipeline integrating Automatic Speech Recognition (Sarvam AI / Whisper), LLM-based semantic parsing, and fuzzy string matching for robust menu item resolution.

Objective 7 (Data Warehousing): To implement a data warehousing pattern with pre-computed daily snapshots enabling sub-second query performance across 90+ days of historical data for dashboard analytics and trend detection.

1.3.4 Project Scope

Domain	Scope	AI/ML Component
Order Management	Full lifecycle: placement, KOT generation, kitchen display, status tracking	Voice ordering (ASR + NLU)
Inventory Management	Ingredient-level BOM tracking, auto-deduction, wastage logging	AI burn-rate analysis, stockout prediction
Billing and Payments	Multi-mode payment, GST, discounts, PDF invoicing	—
AI Analytics	Ask AI chatbot, CrewAI daily brief, Revenue Intelligence	NLP, LLM, MAS, Data Mining
Predictive Analytics	ML-based 7-day revenue forecasting	Polynomial regression

Menu Optimisation	Contribution margins, BCG classification, combo mining	Association analysis, scoring algorithms
Online Orders	Zomato/Swiggy simulation with notification workflow	Realistic data generation
Data Warehousing	DailySummary pre-computed snapshots	Aggregate computation, trend storage
Reporting	Interactive dashboards with 8+ chart types	Time-series visualisation

Table 1.3: Project Scope with AI/ML Component Mapping

1.3.5 Expected Outcomes

- A fully functional, production-deployable restaurant management application validated against realistic data volumes.
- Demonstration of multiple AI/ML techniques (NLP, LLM, MAS, regression, association mining, ASR) integrated within a single domain-specific application.
- A working Big Data analytics pipeline covering data capture → storage → pre-computation → analysis → insight generation → natural language presentation.
- Academic-grade documentation of the analytical methods, algorithms, and evaluation metrics employed.

CHAPTER 2: STATEMENT OF PROBLEM, SYSTEM STUDY, SYSTEM ANALYSIS, SYSTEM DESIGN, APPLICATION AND UTILITY

2.1 Statement of the Problem

2.1.1 Problem Identification

The restaurant industry, despite generating voluminous operational data daily — encompassing orders, payments, inventory movements, wastage records, and multi-channel transactions — overwhelmingly fails to leverage this data for strategic decision-making. Through domain analysis and stakeholder consultation, the following critical problems were identified:

Problem 1 — Data Inaccessibility and Analytical Illiteracy: Restaurant operators, particularly in the SME segment, are not technically equipped to extract useful insights from their POS databases. Raw transactional data resides in database tables, inaccessible to those who need it most. Generating even basic insights such as "What was our most profitable dish last week?" requires SQL expertise or manual spreadsheet extraction — skills absent in 95%+ of restaurant management teams.

Problem 2 — Reactive Inventory Management: Ingredient shortages are typically discovered at the point of failure — when a cashier attempts to place an order and finds that a critical ingredient is exhausted. This reactive approach leads to customer dissatisfaction, revenue loss (estimated at ₹2,000–₹5,000 per stockout event), and emergency procurement at premium prices.

Problem 3 — Absence of Predictive Analytics: No forecasting mechanism exists for anticipating demand fluctuations. Restaurants cannot proactively optimise staffing, procurement quantities, or menu availability based on

predicted demand — leading to either overstaffing/over-procurement (cost waste) or understaffing/stock-outs (revenue loss).

Problem 4 — Unscientific Pricing and Menu Composition: Menu pricing in SME restaurants is typically set based on owner intuition or competitor observation, without reference to actual food cost data, contribution margins, or demand elasticity. Items with low margins and high volumes ("Workhorses") erode profitability, while high-margin items with low demand ("Puzzles" or "Hidden Gems") remain under-promoted.

Problem 5 — No Automated Decision Support: Large restaurant chains employ dedicated data analysts and category managers to produce weekly/monthly performance reviews and action plans. SME restaurants have no equivalent capability — resulting in delayed responses to deteriorating metrics, missed optimisation opportunities, and decision-making by anecdote rather than evidence.

Problem 6 — Multi-Channel Data Fragmentation: The proliferation of third-party food delivery platforms means that restaurants must process orders from multiple channels (dine-in, takeaway, Zomato, Swiggy) without a unified analytical view. Revenue attribution, platform-wise performance comparison, and consolidated inventory impact analysis become impossible without manual data reconciliation.

Problem 7 — Absence of Voice-Enabled Ordering: In busy kitchen environments, staff must interrupt food preparation to manually enter orders via touchscreens. A hands-free, voice-based ordering capability — particularly one supporting Indian languages (Hindi, Gujarati, Tamil, etc.) — would significantly improve operational efficiency.

2.1.2 Problem Statement

"Small-to-medium restaurant enterprises, which collectively generate millions of transactional data points daily, lack the AI-driven analytical infrastructure necessary to transform this raw data into actionable business intelligence — resulting in suboptimal inventory management, uninformed pricing decisions, missed revenue optimisation opportunities, and an inability to proactively address operational risks through predictive analytics and automated decision support."

2.1.3 Proposed Solution — AI and Big Data Analytics Approach

5ive POS addresses the identified problems through a comprehensive AI and Big Data Analytics-driven solution architecture:

Problem	Solution	AI/ML Technique
Data inaccessibility	Ask AI Chatbot — natural language data querying	NLP, LLM (Llama 3.3 70B)
Reactive inventory	AI Inventory Agent with burn-rate analysis and stockout prediction	Multi-Agent System, statistical analysis
No forecasting	ML-based polynomial regression sales forecasting	Supervised learning (regression)
Unscientific pricing	Revenue Intelligence Engine with BCG classification and margin analysis	Data mining, association analysis
No decision support	CrewAI 4-agent daily planning brief	Multi-Agent Systems, LLM reasoning
Multi-channel fragmentation	Unified online order simulation with consolidated analytics	Data integration, automated aggregation

No voice ordering	Voice-to-KOT pipeline with multilingual ASR	ASR (Whisper/Sarvam AI), fuzzy matching

Table 2.1: Problem-to-Solution Mapping with AI/ML Techniques

2.2 Existing System Study

2.2.1 Comparative Analysis of Existing POS Solutions

An analysis of existing restaurant POS systems was conducted to identify analytical capability gaps:

Category A — Traditional POS Systems (Petpooja, Posist, Marg ERP):

- Primarily transactional — billing, order management, basic inventory tracking
- Analytics limited to pre-built static reports (daily/monthly sales, item-wise sales)
- No AI-driven insights, natural language querying, or predictive capabilities
- Inventory management often item-level (not ingredient-level), precluding BOM-based cost analysis
- No multi-agent decision support or automated daily intelligence briefing
- Multi-channel order management requires paid third-party integrations

Category B — Enterprise Systems (Oracle MICROS, Toast POS, Square for Restaurants):

- Full-featured with basic demand forecasting in premium tiers
- AI features limited to simple trend detection — no multi-agent systems or NLP chatbots
- Prohibitively expensive for SMEs (₹15,000–₹50,000/month licensing)
- Require dedicated IT staff for configuration and maintenance
- No voice ordering or multilingual ASR capabilities

Category C — Emerging AI-Enhanced Platforms (limited to R&D/pilot stage):

- Academic prototypes integrating LLMs with restaurant data exist but lack production-grade POS functionality
- No commercially available system combines NLP chatbot, multi-agent decision support, revenue intelligence, voice ordering, and ML forecasting within a single integrated POS platform

2.2.2 Gap Analysis

Feature / Capability	Traditional POS	Enterprise POS	5ive POS
Natural Language Data Querying (NLP)	✗	✗	✓
Multi-Agent AI Daily Brief (MAS)	✗	✗	✓
ML-Based Sales Forecasting	✗	Limited	✓
Revenue Intelligence (BCG, Margins)	✗	✗	✓
Association-Based Combo Recommendation	✗	✗	✓
Algorithmic Price Optimisation	✗	✗	✓
Voice Ordering (Multilingual ASR)	✗	✗	✓
Ingredient-Level BOM Inventory	Partial	✓	✓
Data Warehousing (Pre-computed Snapshots)	✗	✓	✓

Online Order Simulation	✗	✗	✓
Open Source / Customisable	✗	✗	✓

Table 2.2: Comprehensive Gap Analysis of POS Systems

2.3 System Analysis

2.3.1 Feasibility Study

Technical Feasibility: The selected technology stack — FastAPI, React.js, SQLite, CrewAI, Groq Cloud API, Sarvam AI, Whisper, NumPy, and RapidFuzz — consists entirely of mature, well-documented, and freely available (or free-tier accessible) technologies. The Groq Cloud API provides access to the Llama 3.3 70B model and Whisper Large v3 with generous free-tier limits sufficient for development, demonstration, and low-volume production use. The entire stack can execute on a standard developer laptop, requiring no GPU infrastructure for inference (all LLM inference is cloud-hosted).

Economic Feasibility: The system is built entirely on open-source technologies with cloud API dependencies available on free tiers. There are no licensing fees, no proprietary software costs, and no infrastructure rental — making the total development cost limited to developer time. This economic model directly demonstrates that AI-driven business intelligence is achievable for SME restaurants without the capital expenditure traditionally associated with enterprise analytics platforms.

Operational Feasibility: The system prioritises usability for non-technical users. The Ask AI chatbot eliminates technical barriers to data access. The automated daily brief reduces the time managers spend on manual data analysis from an estimated 60–90 minutes to zero (automated delivery). The POS terminal interface follows industry-standard ergonomic patterns. The voice ordering capability enables hands-free interaction in busy kitchen environments.

2.3.2 Requirements Analysis

Functional Requirements:

ID	Requirement	Priority	AI/ML Dependency
FR-01	Create orders with item selection, table assignment, and order type	Critical	Voice ordering (optional)
FR-02	Automatic BOM-based ingredient validation and stock deduction	Critical	—
FR-03	Generate KOTs automatically upon order placement	Critical	—
FR-04	Kitchen Display System with real-time status updates	Critical	—
FR-05	Multi-mode billing (Cash/UPI/Card/Wallet) with GST and discounts	Critical	—
FR-06	PDF invoice generation and email delivery	High	—

FR-07	Ask AI chatbot accepting free-text natural language queries	High	NLP + LLM
FR-08	CrewAI agent execution at 08:00 AM daily, accessible via dashboard	High	MAS + LLM
FR-09	ML-based 7-day sales forecasting with confidence metrics	High	Polynomial regression
FR-10	Revenue Intelligence with margin, velocity, and combo analysis	High	Data mining
FR-11	Voice ordering through ASR with fuzzy menu matching	Medium	ASR + NLU
FR-12	Online order simulation with notification-based approval	Medium	—
FR-13	Interactive reports with charts (sales, peak hours, wastage)	High	Data visualisation
FR-14	Pre-computed daily snapshots for dashboard performance	Medium	Data warehousing

Table 2.3: Functional Requirements with AI/ML Dependencies

Non-Functional Requirements:

ID	Requirement	Target
NFR-01	API response time for POS operations	< 2 seconds
NFR-02	Ask AI response time (rule-based path)	< 500 ms
NFR-03	Ask AI response time (LLM path)	< 3 seconds
NFR-04	Voice-to-KOT pipeline latency	< 5 seconds
NFR-05	Transactional safety for all inventory operations	ACID compliance
NFR-06	Security — JWT authentication, bcrypt password hashing	Industry standard
NFR-07	Database scalability — migration from SQLite to PostgreSQL	Zero business logic changes
NFR-08	Usability — operable by non-technical staff	Minimal training required
NFR-09	ML forecast accuracy	MAPE < 15%

Table 2.4: Non-Functional Requirements

2.4 System Design

2.4.1 High-Level System Architecture

5ive POS implements a **three-tier client-server architecture augmented with an AI analytics layer**:

```
┌──────────────────────────────────────────────────────────────┐
│                PRESENTATION TIER (React.js + Vite)            │
│  POS Terminal │ KDS │ Dashboard │ Reports │ Ask AI Chat       │
│  Inventory │ Menu │ Billing │ Agent Insights │ Voice Order    │
│  Revenue Intelligence │ Online Orders                         │
└──────────────────────┬───────────────────────────────────────┘
                       │  REST API (JSON / HTTP)
┌──────────────────────▼───────────────────────────────────────┐
│                APPLICATION TIER (FastAPI, Python 3.11)        │
│                                                               │
│  ┌──────────────────────────────────────────────────────┐    │
│  │          AI / BIG DATA ANALYTICS LAYER                │    │
│  │                                                       │    │
│  │  ┌──────────────┐  ┌──────────────────────────┐      │    │
│  │  │  Ask AI NLP  │  │  CrewAI Multi-Agent (MAS) │      │    │
│  │  │  (Hybrid     │  │  4 Agents: Inventory,     │      │    │
│  │  │   Rule+LLM)  │  │  Sales, Pricing, CoPilot  │      │    │
│  │  └──────────────┘  └──────────────────────────┘      │    │
│  │                                                       │    │
│  │  ┌──────────────┐  ┌──────────────────────────┐      │    │
│  │  │  ML Forecast │  │  Revenue Intelligence     │      │    │
│  │  │  (NumPy      │  │  (9-Module Analytics:     │      │    │
│  │  │   Poly Reg)  │  │   BCG, Combos, Pricing)   │      │    │
│  │  └──────────────┘  └──────────────────────────┘      │    │
│  │                                                       │    │
│  │  ┌──────────────┐  ┌──────────────────────────┐      │    │
│  │  │  Voice-KOT   │  │  Data Warehouse           │      │    │
│  │  │  (ASR+NLU    │  │  (DailySummary Snapshots)  │    │    │
│  │  │   +FuzzyMatch│  │                           │      │    │
│  │  └──────────────┘  └──────────────────────────┘      │    │
│  └──────────────────────────────────────────────────────┘    │
│                                                               │
│  ┌──────────────────────────────────────────────────────┐    │
│  │           BUSINESS LOGIC LAYER                        │    │
│  │  OrderService · InventoryService · BillingService     │    │
│  │  ReportService · SnapshotService · OnlineOrderSvc     │    │
│  │  EmailService · PDFService · Scheduler                │    │
│  └──────────────────────────────────────────────────────┘    │
└──────────────────────┬───────────────────────────────────────┘
                       │  SQLAlchemy ORM
┌──────────────────────▼───────────────────────────────────────┐
│                    DATA TIER (SQLite)                         │
│  Orders │ OrderItems │ KOTs │ MenuItems │ Categories          │
│  Ingredients │ BOM │ Payments │ Invoices │ DailySummary       │
│  WastageLog │ VoiceOrderLog │ Users │ Restaurants             │
└──────────────────────────────────────────────────────────────┘
```

Figure 2.1: Three-Tier System Architecture with AI Analytics Layer

2.4.2 AI Pipeline Architecture

The AI layer comprises six interconnected analytical pipelines:

```
                    ┌──────────────────┐
                    │  User Query      │
                    │  (Text or Voice) │
                    └──────────┬───────┘
                               │
            ┌──────────────────┼──────────────────┐
            │                  │                  │
            ▼                  ▼                  ▼
┌──────────────┐   ┌──────────────┐   ┌──────────────────┐
│  Ask AI NLP  │   │  Voice-KOT   │   │  Scheduled       │
│  Pipeline    │   │  Pipeline    │   │  Pipelines       │
│              │   │              │   │                  │
│  Rule-Based  │   │  Sarvam ASR  │   │  CrewAI Agents   │
│      ↓       │   │      ↓       │   │  ML Forecasting  │
│  LLM Fallback│   │  Groq Parse  │   │  Daily Snapshot  │
│      ↓       │   │      ↓       │   │  Online Orders   │
│  Query Engine│   │  FuzzyMatch  │   │                  │
│      ↓       │   │      ↓       │   │    (8 AM daily)  │
│  Groq LLM    │   │  Order       │   │                  │
│  Response    │   │  Creation    │   │                  │
└──────────────┘   └──────────────┘   └──────────────────┘
            │                  │                  │
            └──────────────────┼──────────────────┘
                               ▼
                  ┌──────────────────────┐
                  │  Revenue Intelligence │
                  │  (9-Module Engine)    │
                  │                       │
                  │  Margins │ Profitability│
                  │  BCG     │ Hidden Gems  │
                  │  Combos  │ Upsells      │
                  │  Pricing │ Inv Signals  │
                  └──────────────────────┘
```

Figure 2.2: AI Pipeline Architecture

2.4.3 Database Design

The database comprises **15 normalised tables** designed to Third Normal Form (3NF):

Table	Primary Key	Records	Purpose
`restaurants`	restaurant_id	1	Restaurant configuration and profile
`users`	user_id	2+	Staff accounts with authentication data
`categories`	category_id	5+	Menu item categories
`menu_items`	item_id	10+	Dishes with prices and availability
`ingredients`	ingredient_id	15+	Raw materials with stock and expiry
`bill_of_materials`	bom_id	30+	Recipe mapping: menu item → ingredients

`orders`	order_id	7,000+	Order headers with type, status, timestamps
`order_items`	order_item_id	12,000+	Line items within each order
`kots`	kot_id	12,000+	Kitchen Order Tickets
`payments`	payment_id	6,500+	Payment transactions
`invoices`	invoice_id	—	Generated invoice records
`wastage_log`	wastage_id	—	Food waste tracking records
`daily_summary`	summary_id	90	Pre-computed daily aggregate metrics
`voice_order_log`	log_id	—	Voice order transcripts and parsed data
`inventory_transactions`	txn_id	—	Audit trail for stock movements

Table 2.5: Database Schema Overview

2.4.4 Entity-Relationship Model

The core relationships in the normalised schema are:

- One **Restaurant** has many **Users**, **Categories**, and **Ingredients** (1:N)
- One **Category** contains many **Menu Items** (1:N)
- One **Menu Item** maps to many **Ingredients** through **Bill of Materials** (M:N junction)
- One **Order** contains many **Order Items** and many **KOTs** (1:N)
- One **Order** has one **Payment** and one **Invoice** (1:1)
- **Wastage Log** entries reference **Ingredients** (N:1)
- **Daily Summary** stores one aggregated record per calendar day (1:1 with date)
- **Voice Order Log** records ASR transcripts and parsed JSON for each voice interaction

2.4.5 API Design

The backend exposes a RESTful API through 11 route groups:

Route Group	Base Path	Key Operations	AI Component
Authentication	`/api/auth`	Register, Login, Get Current User	—
Menu	`/api/menu`	CRUD for categories and items	—
Inventory	`/api/inventory`	Ingredients, stock management, wastage	—
Orders	`/api/orders`	Create, list, update status, cancel	—
KDS	`/api/kds`	Kitchen ticket management	—
Billing	`/api/billing`	Calculate, process payment, invoice	—
Reports	`/api/reports`	Sales, peak hours, wastage, cost analysis, forecasting	ML Forecast
AI / Ask AI	`/api/ai`	Natural language query endpoint	NLP + LLM

Agents	`/api/agents`	Trigger and retrieve CrewAI brief	MAS + LLM
Revenue Intel	`/api/revenue-intelligence`	Margins, combos, velocity, pricing	Data Mining
Voice Bot	`/api/voice-bot`	Voice order processing and confirmation	ASR + NLU

Table 2.6: API Route Groups with AI Component Mapping

2.5 Application and Utility

2.5.1 Operational Utility

Front-of-House Operations:

- POS Terminal with item search, category filtering, and table assignment
- Multi-order type support (dine-in, takeaway, delivery)
- Voice-based order creation for hands-free environments
- Real-time online order notifications (Zomato/Swiggy) with accept/reject workflow

Kitchen Operations:

- Kitchen Display System with colour-coded urgency indicators
- KOT-based order tracking (Placed → Preparing → Ready → Served)
- Voice-generated KOTs for orders placed through the ASR pipeline

Management and Analytics:

- Ask AI chatbot for instant natural language data access
- AI-generated daily planning brief with prioritised action items
- Revenue Intelligence dashboard with BCG classification and combo recommendations
- Interactive reports with 8+ chart types (line, bar, pie, heat map, area)
- ML-powered 7-day revenue forecast with confidence metrics

2.5.2 Strategic and Analytical Utility

Utility Area	Mechanism	Expected Business Impact
Revenue Optimisation	Pricing Agent + Revenue Intelligence margin analysis	+5–15% revenue through informed pricing
Waste Reduction	Burn-rate analysis + expiry alerts + predictive ordering	-10–20% reduction in food cost waste
Demand Planning	ML forecasting + daily brief + peak hour analysis	Improved staffing and procurement decisions
Menu Engineering	BCG classification + combo mining + upsell scripting	Optimised menu composition for profitability
Data Access Democratisation	NLP chatbot — no SQL/technical knowledge required	Reduced time-to-insight from hours to seconds
Decision Support	4-agent daily brief — delivered automatically at 8 AM	Proactive rather than reactive management

Table 2.7: Strategic Utility and Expected Business Impact

CHAPTER 3: METHODOLOGY AND TECHNICAL BACKGROUND

3.1 Development Methodology

3.1.1 Iterative and Incremental Development

The project followed an **Iterative and Incremental Development** methodology, structured across four major increments:

Increment 1 — POS Foundation (Weeks 1–3):

- Database schema design and SQLAlchemy ORM model creation
- FastAPI backend with CRUD APIs for menu, orders, inventory, and authentication
- React.js frontend with POS Terminal and Kitchen Display System
- JWT-based authentication and role-based access control

Increment 2 — Business Logic & Reporting (Weeks 4–6):

- Bill of Materials (BOM) system with automatic inventory deduction
- Transactional rollback mechanism on order cancellation
- Multi-mode billing (GST, discounts, UPI/Cash/Card/Wallet)
- PDF invoice generation via ReportLab
- Reports dashboard with Recharts interactive visualisations
- Daily snapshot service with APScheduler

Increment 3 — AI/ML Integration (Weeks 7–10):

- Ask AI chatbot with hybrid NLP pipeline (rule-based + LLM)
- CrewAI Multi-Agent System with four specialised agents
- Polynomial regression sales forecasting (NumPy)
- Online order simulation (Zomato/Swiggy)
- Automated email briefing via SMTP

Increment 4 — Advanced Analytics & Voice (Weeks 11–14):

- Revenue Intelligence & Menu Optimisation Engine (9 modules)
- Voice-to-KOT pipeline (Sarvam AI ASR + Groq Whisper + RapidFuzz)
- AI query enhancements for complex revenue metrics
- Testing, performance evaluation, and documentation

3.1.2 Version Control

Git with a feature-branch workflow was employed throughout development. Each major feature was developed in an isolated branch and merged upon successful testing.

3.2 Technology Stack

3.2.1 Backend Technologies

Technology	Version	Purpose
FastAPI	Python 3.11	Async REST API framework
SQLAlchemy	2.0	ORM for database abstraction
SQLite	3.x	Embedded relational database
APScheduler	—	Background task scheduling
ReportLab	—	PDF invoice generation
python-jose	—	JWT authentication
Pydantic	2.x	Request/response schema validation
bcrypt	—	Password hashing

Table 3.1: Backend Technology Stack

3.2.2 AI/ML Technologies

Technology	Purpose	Component
Groq Cloud API	LLM inference (Llama 3.3 70B Versatile)	Ask AI, CrewAI, Voice parse
CrewAI	Multi-Agent System orchestration	Daily agent brief
NumPy	Numerical computing, polynomial regression	Sales forecasting
Sarvam AI	Multilingual ASR (Saaras v3 model)	Voice ordering (primary STT)
Groq Whisper	ASR fallback (Whisper Large v3)	Voice ordering (fallback STT)
RapidFuzz	Fuzzy string matching	Voice order menu matching
Groq JSON Mode	Structured LLM output	Intent classification

Table 3.2: AI/ML Technology Stack

3.2.3 Frontend Technologies

Technology	Purpose
React.js 18	Component-based SPA framework
Vite	Fast HMR build tooling
Recharts	Interactive data visualisation (Line, Bar, Pie, Area, Heat Map)
Axios	HTTP client with JWT interceptors
React Router v6	Client-side routing

Table 3.3: Frontend Technology Stack

3.3 AI and Machine Learning Techniques — Detailed Analysis

3.3.1 Natural Language Processing — Hybrid Intent Classification

The Ask AI module implements a **novel hybrid three-layer NLP pipeline** that balances speed, accuracy, and cost:

Layer 1: Rule-Based Intent Classification (Fast Path)

The rule-based classifier employs a **weighted keyword pattern matching** algorithm. For each incoming user query q:

1. The query is normalised to lowercase.
2. An analytical keyword set A is checked to determine if the query is analytical vs. transactional.
3. For each intent category $c \in \{sales_query, inventory_query, order_status, menu_info, wastage_query, create_order, revenue_intel, general\}$, the classifier searches for keyword patterns K_c associated with that category.
4. A confidence score σ_c is assigned based on keyword matches.
5. The intent with the highest σ_c exceeding the confidence threshold $\theta = 0.6$ is returned.

Intent Category	Representative Keywords	Confidence
`sales_query`	revenue, earn, income, sales, profit, growth, forecast, AOV	0.9
`inventory_query`	stock, ingredient, low stock, reorder, expiry, shelf life	0.9
`revenue_intel`	margin, profitability, popular, velocity, combo, upsell, BCG, risk, food cost	0.95
`order_status`	order status, pending, KOT, kitchen, active order	0.85
`menu_info`	menu, dish, price, available	0.8
`wastage_query`	wastage, waste, damaged	0.85
`create_order`	add, order, i want, get me, place order	0.9

Table 3.4: Intent Categories with Keywords and Confidence Scores

This rule-based layer handles approximately 80% of typical restaurant queries with sub-100ms latency and zero API cost.

Layer 2: LLM-Based Fallback Classification (Semantic Path)

When the rule-based classifier returns a confidence score below θ or assigns the `general` category, the query is forwarded to the Groq LLM API with a carefully engineered **few-shot classification prompt**. The prompt includes:

- A comprehensive intent taxonomy with descriptions and examples
- Time period extraction rules with examples
- Instruction to output structured JSON with `intent_type`, `confidence`, `entities`, and `needs_data` fields
- JSON mode enforcement (`response_format: {"type": "json_object"}`) to guarantee parseable output

The LLM temperature is set to 0.05 (near-deterministic) for classification accuracy.

Layer 3: Query Engine + LLM Response Generation

Once intent is classified, the system:

1. Routes to the appropriate **Query Engine** function, which translates the intent into SQLAlchemy ORM queries.
2. Retrieves structured data from the database.
3. Constructs a rich **system prompt** (68+ lines) defining the AI's personality, formatting rules, chart generation instructions, and database schema context.
4. Injects the retrieved data as a JSON block into the LLM prompt.
5. Calls Groq's Llama 3.3 70B with the enriched prompt and conversation history.
6. Parses the response for embedded `__chart__` data blocks for frontend visualisation rendering.

Named Entity Extraction:

The NLP pipeline performs named entity extraction across four entity types:

Entity Type	Pattern	Example
Time Period	Keyword matching + regex	"today", "this week", "last 30 days"
Order Number	Regex: `ORD-\d{8}-\d{4}`	ORD-20240315-0042
Item Name	Quoted strings, fuzzy matching	"Margherita Pizza"
Quantity	Regex: `(\d+)\s+([a-z\s]+)`	"2 burgers and 1 coke"

Table 3.5: Named Entity Types in the NLP Pipeline

3.3.2 Multi-Agent System (MAS) — CrewAI Architecture

The Multi-Agent System implements a **Sequential Crew Pattern** with four autonomous AI agents:

Agent Configuration:

Agent	Role	Goal	Backstory
Inventory Agent	Senior Inventory Analyst	Analyse stock levels, predict shortages, minimise wastage	Expert inventory manager with 15 years of supply chain experience
Sales Agent	Revenue & Sales Strategist	Analyse sales performance, identify patterns, find growth	Seasoned analyst who has optimised revenue for 100+ restaurants
Pricing Agent	Pricing Optimisation Specialist	Optimise menu pricing using cost analysis and demand	Menu engineering expert in pricing psychology and profit optimisation
Co-Pilot Agent	Restaurant AI Co-Pilot	Synthesise all insights into concise daily planning brief	Trusted AI advisor who prioritises ruthlessly

Table 3.6: CrewAI Agent Definitions

Novel Data Pre-Fetching Strategy:

A critical design decision in the MAS implementation was the adoption of a **data pre-fetching injection strategy** rather than the conventional LLM tool-calling approach. The rationale:

Aspect	Tool-Calling Approach	Pre-Fetching Approach (5ive POS)

Data Access	LLM generates function calls → API executes	Data pre-fetched before agents start
Reliability	Risk of malformed calls, infinite loops	100% deterministic data delivery
Latency	Multiple round-trips per agent	Zero round-trips during execution
Token Cost	Additional tokens for tool call prompts	Slightly larger initial prompt
Complexity	Requires tool registration, error handling	Simple prompt injection

Table 3.7: Comparison of Agent Data Access Strategies

The pre-fetching function `_fetch_all_data()` queries all sales summaries, inventory levels, peak hours, item performance, wastage reports, ingredient usage, and cost analysis data — converting each dataset to JSON and injecting it directly into agent task prompts. This ensures that LLM agents focus exclusively on analysis and recommendation generation, with no database interaction required.

Crew Execution:

The agents execute sequentially: Inventory → Sales → Pricing → Co-Pilot. The Co-Pilot agent receives the output of all three preceding agents as context and synthesises a unified daily brief structured as:

- 🎯 Top 3 Priorities
- 📦 Inventory Actions
- 💰 Sales Insights
- 🏷️ Pricing Suggestions
- 📊 Tomorrow Forecast
- ⚠️ Risk Alerts

3.3.3 Machine Learning — Polynomial Regression Forecasting

Algorithm: Polynomial Regression (Degree-2), implemented using NumPy.

Mathematical Formulation:

The model fits a quadratic polynomial to historical daily revenue data:

$\hat{y} = \beta_2 x^2 + \beta_1 x + \beta_0$

Where:

- $\hat{y}$ = predicted daily revenue (₹)
- x = day index (0, 1, 2, ..., n)
- $\beta_2, \beta_1, \beta_0$ = fitted coefficients computed using `numpy.polyfit(x, y, deg=2)`

Implementation:

```
# Actual implementation in report_service.py
coefficients = numpy.polyfit(x_values, y_revenue, deg=2)
polynomial = numpy.poly1d(coefficients)
forecast = polynomial(future_x_values)  # Predict days n+1 to n+7
```

Model Evaluation Metrics:

Metric	Formula	Purpose

R^2 (Coefficient of Determination)	1 − (SS_res / SS_tot)	Goodness of fit (0–1 scale)
MAE (Mean Absolute Error)	$(1/n)\ \Sigma\lvert y_i - \hat{y}_i\rvert$	Average prediction error (₹)
MAPE (Mean Absolute Percentage Error)	$(1/n)\ \Sigma\lvert y_i - \hat{y}_i\rvert/y_i \times 100$	Normalised error (%)

Justification for Polynomial Regression over Advanced Models:

Criterion	Polynomial Regression	ARIMA	LSTM
Interpretability	High (coefficients visible)	Medium	Low (black box)
Implementation complexity	Low (3 lines of NumPy)	High	Very High
Dependencies	NumPy only	statsmodels	TensorFlow/PyTorch
Training data required	30–90 days sufficient	100+ days	500+ days
Non-linear trend capture	Yes (quadratic)	Limited	Excellent
Academic clarity	Excellent	Good	Poor
Accuracy for 7-day horizon	Adequate (MAPE 8–14%)	Good	Overkill

Table 3.8: Comparison of Forecasting Techniques

For a 7-day forecast horizon over 30–90 days of daily restaurant revenue data, polynomial regression provides the optimal balance of accuracy, interpretability, and implementation simplicity — making it well-suited for both production deployment and academic demonstration.

3.3.4 Revenue Intelligence — Association Rule Mining and BCG Classification

The Revenue Intelligence Engine (`revenue_intelligence_service.py`) implements nine analytical modules:

Module 1: Contribution Margin Calculation

For each menu item *i*:

Margin_i = Selling Price_i − Food Cost_i (from BOM)

Margin %_i = (Margin_i / Selling Price_i) × 100

Items are classified into margin tiers: **High** (≥60%), **Medium** (40–59%), **Low** (<40%).

Module 2: Item-Level Profitability Analysis

Total profit contribution, revenue, units sold, and profit-per-unit are computed for each item over a configurable time window. Items are ranked by absolute profit contribution and assigned a percentage share of total profit.

Module 3: Sales Velocity & BCG-Style Classification

Velocity score (0–100) is computed as a normalised measure of daily sales frequency. Items are classified using a **BCG Growth-Share Matrix** adaptation:

Quadrant	Velocity	Margin	Strategy
Star ⭐	≥ Average	≥ Average	Keep promoting — high demand, high profit

Workhorse 🐂	≥ Average	< Average	Optimise cost or raise price — sells well but thin margins
Puzzle 🧩	< Average	≥ Average	Needs promotion — great margins, poor demand
Dog 🐕	< Average	< Average	Consider removing — poor on both dimensions

Table 3.9: BCG-Style Menu Item Classification

Module 4: Under-Promoted High-Margin Detection ("Hidden Gems")

Items with margin ≥ 60% but below-average velocity are flagged as Hidden Gems — high profit potential items that are under-promoted. The engine calculates potential daily revenue if velocity were brought to average.

Module 5: Low-Margin High-Volume Risk Detection

Items with margin < 45% and above-average velocity are flagged as profitability risks — products actively eroding margins due to high demand at low margins.

Module 6: Combo Recommendation via Association Analysis

The engine implements a simplified **market basket analysis** approach:

1. All order records within the analysis window are grouped by order ID.
2. For each order containing ≥2 distinct items, all item pairs are enumerated.
3. **Co-occurrence counts** are computed using a Counter data structure.
4. Pairs appearing in ≥ 3% of orders (minimum support) and ≥ 5 absolute occurrences are selected.
5. A **12% discount** is applied to suggest combo pricing.
6. Projected monthly revenue and orders are extrapolated.

This approach is conceptually aligned with the **Apriori algorithm** for association rule mining, simplified for the specific domain constraint of two-item combos.

Module 7: Smart Upsell Prioritisation

For each popular item, the engine identifies a higher-margin item suitable for upselling, computes the incremental profit impact assuming a 15% conversion rate, and generates a specific upsell script.

Module 8: Algorithmic Price Optimisation

Price recommendations are computed using the following rules:

Current State	Recommendation	Calculation
Low margin + High velocity	Increase price	Min(15%, avg_margin − item_margin)
High margin + Low velocity (<60% avg)	Decrease price	Min(10%, (avg_margin − margin) × -0.3)
Otherwise	Hold	—

Projected monthly revenue is computed for each recommendation.

Module 9: Inventory-Linked Performance Signals

Cross-references item profitability with ingredient stock levels. Flags critical situations where high-profit items are at risk of stockout — computing servings remaining, daily usage, and days until stockout.

3.3.5 Voice-to-KOT Pipeline — Automatic Speech Recognition and Natural Language Understanding

The Voice ordering system implements a **three-stage pipeline**:

Stage 1: Automatic Speech Recognition (ASR)

A **multi-provider fallback architecture** ensures maximum reliability:

Priority	Provider	Model	Language Support	Latency
1 (Primary)	**Sarvam AI**	Saaras v3	12+ Indian languages → English	< 1 second
2 (Fallback)	**Groq Cloud**	Whisper Large v3	50+ languages	~2 seconds
3 (Local)	**OpenAI Whisper**	Base (local)	English	~3–5 seconds

Table 3.10: ASR Provider Hierarchy

Audio pre-processing includes conversion to 16kHz mono WAV using FFmpeg. A **hallucination detection** system filters known Whisper failure patterns (e.g., "Thank you for watching", "Subscribe") that occur with silent or noisy audio input.

Stage 2: LLM-Based Semantic Order Parsing

The raw transcript is sent to an LLM (Sarvam AI primary, Groq fallback) with a carefully engineered prompt that includes:

- The complete restaurant menu with categories and prices
- Parsing rules for quantities ("a pizza" = 1, "couple of" = 2, etc.)
- Special instruction extraction (e.g., "extra spicy", "no onion")
- Menu item name matching instruction

The LLM returns a structured JSON array of parsed items.

Stage 3: Fuzzy String Matching (RapidFuzz)

Parsed item names are matched against the actual menu using **token sort ratio** as the similarity metric:

Confidence Score	Label	Action
≥ 85%	HIGH	Auto-accept match
65–84%	MEDIUM	Accept with visual confirmation
< 65%	LOW	Flag for manual review

Table 3.11: Fuzzy Match Confidence Tiers

3.4 Big Data Analytics Techniques

3.4.1 Data Warehousing — DailySummary Snapshot Service

The `snapshot_service.py` implements a lightweight data warehousing pattern inspired by star schema architectures. Each row in the `daily_summary` table stores pre-computed aggregate metrics for a single calendar day:

Metric	Computation
Total Revenue	SUM(order.total_amount) WHERE status ≠ CANCELLED
Order Count	COUNT(orders)
Average Order Value	Revenue / Order Count
Top-Selling Item	MAX(SUM(order_items.quantity)) GROUP BY item
Payment Mode Distribution	JSON aggregate of payment counts by mode
Peak Hour	MODE(HOUR(order.created_at))
Online vs Offline Split	COUNT WHERE source IN ('zomato', 'swiggy') / Total
Wastage Cost	SUM(wastage_log.cost)

Table 3.12: DailySummary Pre-Computed Metrics

This pattern reduces query time for historical analytics from O(n) aggregation over all orders to O(1) lookup on the summary table — enabling sub-second dashboard loading across 90+ days of data.

3.4.2 Data Pipeline Architecture

The data flow from capture to insight follows a structured pipeline:

```
Raw Transactions            →  Normalised Storage (3NF)
  (Orders, Payments,             (SQLite via SQLAlchemy)
   Inventory Movements)

Normalised Storage          →  Pre-Computed Aggregates
  (DailySummary Snapshots)       (nightly APScheduler job)

Pre-Computed Aggregates     →  AI Analysis Layer
  (Agent data pre-fetching,      (CrewAI, Revenue Intelligence)
   Dashboard data)

AI Analysis Layer           →  Natural Language Output
  (LLM response generation,      (Ask AI chatbot, Email brief)
   Chart data serialisation)
```

Figure 3.1: Data-to-Insight Pipeline

3.4.3 Privacy-Preserving Synthetic Data Simulation

A critical challenge when developing Big Data Analytics systems for enterprise domains is acquiring statistically realistic historical data without violating data privacy laws (such as the DPDP Act) or corporate confidentiality. Real restaurant POS data contains highly sensitive business intelligence, including exact profit margins, ingredient supply costs, daily revenue figures, and customer Personally Identifiable Information (PII).

To scientifically validate the ML Sales Forecasting and the Revenue Intelligence Engine (which require 90 to 180 days of historical data to function), a **Privacy-Preserving Synthetic Data Simulator** was engineered. This simulator

algorithmically generates transactional data that adheres to real-world statistical distributions rather than random noise:

- **Poisson Distribution:** Used to simulate order arrival times, accurately creating realistic traffic clusters during lunch (12:00–14:00) and dinner (19:00–22:00) peak hours.
- **Pareto Principle (80/20 Rule):** Applied to menu item popularity, ensuring that approximately 20% of the menu generates 80% of the sales volume (accurately simulating 'Star' and 'Workhorse' items for the BCG matrix).
- **Realistic Cost Margins:** The data generator calculates food costs using the Bill of Materials (BOM) to ensure Contribution Margins strictly remain within the realistic Indian restaurant economic boundaries of 40–60%.

The entire Big Data pipeline — from the FastAPI backend through the SQLAlchemy ORM to the CrewAI agents — is fundamentally **data-agnostic**. The algorithms process the synthetic data exactly identically to how they would process real production data. This approach demonstrates enterprise-grade system validation while strictly adhering to data privacy and security principles.

3.5 Tools and Development Environment

Tool	Purpose
VS Code	Primary IDE for full-stack development
Postman	API testing and endpoint documentation
Git + GitHub	Version control with feature-branch workflow
SQLite Browser	Database inspection and query testing
Groq Console	LLM API key management and usage monitoring
Sarvam AI Dashboard	ASR API key management
Node.js 18	JavaScript runtime for frontend build
Python 3.11	Backend runtime environment
pip / npm	Package management
FFmpeg	Audio format conversion for voice pipeline

Table 3.13: Development Tools

CHAPTER 4: DETAILS OF ANALYSIS

4.1 Functional Requirements Analysis

4.1.1 Use Case Analysis

The 5ive POS system identifies four primary actors:

- **Cashier:** Operates the POS Terminal, places orders, processes payments, generates invoices.
- **Kitchen Staff:** Views and updates KOTs on the Kitchen Display System; may use voice commands.
- **Restaurant Manager (Admin):** Manages menu, inventory, reports, AI insights, Revenue Intelligence, and online order approvals.

- **System (Automated):** APScheduler tasks that generate online orders, trigger the daily AI brief, compute daily snapshots, and run ML forecasts.

Primary Use Cases:

UC-01: Place Order (Manual)

- Actor: Cashier
- Precondition: Authenticated; menu items with BOM configured.
- Flow: Cashier selects items → system validates BOM stock → order created → KOT dispatched → inventory deducted.
- Postcondition: Order in "Placed" status; KOT visible on KDS; inventory reduced.

UC-02: Place Order (Voice)

- Actor: Kitchen Staff / Customer
- Precondition: Voice pipeline loaded; microphone active.
- Flow: User speaks order → Sarvam AI / Whisper transcribes → Groq LLM parses → RapidFuzz matches items → user confirms → order created.
- Postcondition: Voice order log saved; order created with matched items.

UC-03: Process Payment and Invoice

- Actor: Cashier
- Precondition: Order in "Served" status.
- Flow: Billing calculates total + GST + discount → payment mode selected → payment recorded → PDF invoice generated.
- Postcondition: Order "Completed"; payment and invoice saved.

UC-04: Query via Ask AI

- Actor: Restaurant Manager
- Precondition: Authenticated; historical data exists.
- Flow: Manager types query → hybrid NLP classifies intent → database queried → Groq LLM formats response with charts → response displayed.
- Postcondition: Natural language data response with optional embedded chart.

UC-05: View Daily AI Brief

- Actor: Restaurant Manager
- Precondition: CrewAI executed (8 AM auto or manual trigger).
- Flow: Manager opens Agent Insights → system retrieves brief → displays structured sections for Inventory, Sales, Pricing, and Co-Pilot recommendations.
- Postcondition: Manager has prioritised daily action list.

UC-06: Analyse Revenue Intelligence

- Actor: Restaurant Manager
- Precondition: Menu items with BOM; sales data exists.
- Flow: Manager opens Revenue Intelligence → system computes all 9 modules → displays margins, BCG tiers, combos, upsells, pricing recommendations.
- Postcondition: Manager has data-driven menu optimisation insights.

UC-07: Process Online Order

- Actor: Restaurant Manager
- Precondition: Online order simulation running.

- Flow: Notification popup appears → manager reviews → Accept/Reject → if accepted, order enters POS workflow.
- Postcondition: Accepted orders integrated into POS.

4.2 Non-Functional Requirements Analysis

4.2.1 Performance Analysis

Under single-user testing conditions, the following response times were measured:

Operation	Average Response Time	Target	Status
Order creation (manual)	~180 ms	< 2s	✅ Met
Billing calculation	~90 ms	< 2s	✅ Met
Ask AI — rule-based path	~350 ms	< 500ms	✅ Met
Ask AI — LLM response generation	~2.1 s	< 3s	✅ Met
Revenue Intelligence full report	~800 ms	< 2s	✅ Met
ML sales forecasting	~120 ms	< 1s	✅ Met
Daily AI brief generation (CrewAI)	~45–90 s	< 120s	✅ Met
Dashboard data load	~250 ms	< 1s	✅ Met
Voice-to-KOT pipeline (end-to-end)	~3.2 s	< 5s	✅ Met
Voice — Sarvam AI STT only	~800 ms	—	—
Voice — Groq Whisper STT only	~1.8 s	—	—
Voice — RapidFuzz matching	~5 ms	—	—

Table 4.1: System Performance Benchmarks

The LLM response time of ~2.1 seconds is primarily attributable to Groq Cloud API network latency, not local processing. All local operations (query execution, data formatting, rule-based classification) complete within 500 ms.

The CrewAI daily brief (45–90 seconds) executes as a background job and does not impact user-facing responsiveness.

4.2.2 Data Volume Analysis

The system was tested against a synthetic dataset representing a realistic 90-day restaurant operation:

Data Category	Record Count	Data Size
Total orders	7,000+	~2 MB
Order items	12,000+	~3 MB
Kitchen Order Tickets	12,000+	~3 MB

Payment records	6,500+	~1.5 MB
Daily summary records	90	~50 KB
Menu items	10+	~5 KB
Ingredient records	15+	~8 KB
BOM mappings	30+	~3 KB
Total database size	—	**~14.8 MB**

Table 4.2: Dataset Dimensions

The system maintains sub-second response times across all dashboard and report queries even with this data volume, attributable to:

1. Pre-computed daily summary records eliminating real-time aggregation.
2. Indexed foreign keys on high-frequency join columns.
3. SQLAlchemy query optimisation with selective eager loading.

4.2.3 Scalability Assessment

Scalability Dimension	Current Capacity	Scaling Path
Database engine	SQLite (single-writer)	PostgreSQL (concurrent R/W)
LLM inference	Groq free tier (30 RPM)	Paid tier or self-hosted LLM
Data volume	~1 year of single-restaurant data	Partitioned tables, time-series DB
Multi-tenancy	Single restaurant	Tenant-based schema isolation

Table 4.3: Scalability Assessment

4.3 Database and Data Engineering Analysis

4.3.1 Normalisation Analysis

The database schema is designed to **Third Normal Form (3NF)**:

- **1NF:** All tables have a primary key; no repeating groups; atomic values in all cells.
- **2NF:** No partial dependencies on composite keys (all tables use single-column primary keys).
- **3NF:** No transitive dependencies — every non-key attribute depends only on the primary key.

The `bill_of_materials` table is a classic **junction table** (many-to-many relationship between `menu_items` and `ingredients`), correctly decomposed to eliminate data redundancy.

4.3.2 Data Warehousing Pattern — DailySummary

The `daily_summary` table implements a **fact table pattern** from dimensional modelling. Each row is a fact record with one row per calendar day, containing pre-computed aggregate measures:

Measure	Derivation
Total Revenue	Real-time SUM of completed order amounts

Order Count	Count of non-cancelled orders
Avg Order Value	Revenue / Order Count
Top Item	Item with highest quantity sold
Payment Distribution	JSON: {cash: n, upi: n, card: n, wallet: n}
Peak Hour	Hour with maximum order count
Online/Offline Split	Percentage of orders from delivery platforms
Wastage Cost	Sum of wastage log entries

This pattern reduces dashboard query complexity from **O(n)** (aggregation over all orders) to **O(1)** (direct lookup), providing consistent sub-250ms dashboard loading regardless of data growth.

4.3.3 Transaction Safety — ACID Compliance Analysis

All inventory-modifying operations implement explicit SQLAlchemy transactions:

Order Placement Transaction:

1. BEGIN TRANSACTION
2. For each order item: retrieve BOM → check ingredient stock → deduct quantities → log InventoryTransaction
3. Create Order record → Create OrderItem records → Create KOT records
4. COMMIT (or ROLLBACK on any failure)

Order Cancellation Transaction:

1. BEGIN TRANSACTION
2. Retrieve all InventoryTransaction records for the order
3. Reverse each deduction (add stock back)
4. Update order status to CANCELLED
5. COMMIT

This ensures **atomicity** (all-or-nothing), **consistency** (no partial deductions), **isolation** (concurrent orders cannot interfere), and **durability** (committed changes survive system failure).

4.4 AI Layer — Detailed Analytical Evaluation

4.4.1 Ask AI — Intent Classification Accuracy

The hybrid NLP pipeline was evaluated against a representative set of natural language queries:

Intent Category	Sample Query	Classification Method	Accuracy Assessment
`sales_query`	"How much did we earn this week?"	Rule-based (σ=0.9)	High — keywords sufficient
`inventory_query`	"Which ingredients are expiring?"	Rule-based (σ=0.9)	High — keywords sufficient

`revenue_intel`	"What are our most profitable items?"	Rule-based (σ=0.95)	High — revenue keywords prioritised
`order_status`	"What is the status of ORD-12345678-0001?"	Rule-based (σ=0.85)	High — pattern match on order number
`menu_info`	"What's the price of Paneer Tikka?"	Rule-based (σ=0.8)	High — keyword match
`wastage_query`	"How much did we waste this month?"	Rule-based (σ=0.85)	High — keyword match
`create_order`	"Add 2 burgers and a coke to table 5"	Rule-based (σ=0.9)	Medium — requires entity extraction
Complex query	"Compare this week's revenue with last week"	LLM fallback	High — LLM handles temporal comparison
Ambiguous query	"How is the restaurant doing?"	LLM fallback	High — LLM infers sales intent

Table 4.4: Ask AI Intent Classification Evaluation

Key Observation: The rule-based classifier successfully handles ~80% of typical restaurant queries with sub-100ms latency. The LLM fallback is invoked only for genuinely complex or ambiguous queries, resulting in optimal cost efficiency (minimised API calls) and latency (fast-path for common queries).

4.4.2 Ask AI — Response Quality Analysis

The Groq LLM response generation was evaluated on the following dimensions:

Dimension	Assessment	Evidence
Data Accuracy	Excellent	LLM uses only injected data; no hallucinated numbers observed
Formatting	Excellent	Consistent Markdown with tables, bold numbers, emoji indicators
Chart Generation	Good	Correctly generates `__chart__` blocks for multi-point data
Conversational Context	Good	Maintains 6-message history for follow-up queries
Fallback Behaviour	Good	Template-based responses when LLM is unavailable

Table 4.5: LLM Response Quality Assessment

4.4.3 CrewAI Multi-Agent System — Output Quality Analysis

The Multi-Agent System was evaluated qualitatively:

Agent	Output Assessment	Strengths	Limitations
Inventory Agent	Accurately identifies low-stock items; computes stockout ETAs; FIFO recommendations are relevant	Data-specific, actionable alerts	Requires Groq API; ~15s per agent

Sales Agent	Correctly identifies top performers and trends; peak hour analysis aligned with data	Revenue-focused with growth metrics	Formatting varies across runs
Pricing Agent	Recommendations logically derived from cost and demand data	Specific ₹ amounts suggested	Conservative (max 10–15% changes)
Co-Pilot Agent	Well-structured daily brief; priorities ranked logically	Concise, actionable, comprehensive	Occasionally verbose beyond 350 words

Table 4.6: CrewAI Agent Output Quality

Reliability: The system maintains consistent output quality across multiple executions with the same data. The data pre-fetching strategy eliminates the non-determinism associated with LLM tool-calling, ensuring agents always receive complete and correctly formatted data.

4.4.4 ML Forecasting — Accuracy Analysis

The polynomial regression model was evaluated using a **train-test split** methodology:

- **Training data:** Days 1–83 (83 daily revenue records)
- **Test data:** Days 84–90 (7 daily revenue records)

Metric	Observed Range	Interpretation
R^2 (Training)	0.82 – 0.91	Good fit — model captures ~85% of variance
MAE (Test)	₹1,200 – ₹2,800 per day	Acceptable for planning purposes
MAPE (Test)	8% – 14%	Within acceptable range for 7-day forecasts

Table 4.7: ML Forecast Evaluation Metrics

Performance by Data Pattern:

- **Steady-state periods:** MAPE 5–8% (excellent)
- **Growth/decline trends:** MAPE 8–12% (good, captured by quadratic term)
- **Anomalous days (holidays/events):** MAPE 15–25% (poor — expected limitation)

The quadratic model effectively captures smooth non-linear trends but cannot account for discrete events (festivals, weather disruptions) or day-of-week seasonality. These acknowledged limitations are addressed in the Recommendations section.

4.5 Revenue Intelligence Engine — Algorithmic Analysis

4.5.1 Contribution Margin Module Performance

The margin calculation engine was validated against manual computations:

Menu Item	Selling Price (₹)	Food Cost via BOM (₹)	System Margin (₹)	System Margin (%)	Manual Verification
Margherita Pizza	350	98	252	72.0%	✓ Correct

Paneer Tikka	280	72	208	74.3%	✓ Correct
Chicken Burger	220	105	115	52.3%	✓ Correct
Cold Coffee	150	32	118	78.7%	✓ Correct

Table 4.8: Margin Calculation Verification (Illustrative)

4.5.2 BCG Classification Analysis

The BCG-style classification was applied to the menu, yielding the following distribution:

Tier	Count	Typical Characteristics	Strategic Action
Star	~25%	Top-selling items with healthy margins	Maintain visibility; protect supply chain
Workhorse	~30%	High-demand items with low margins	Optimise ingredient cost or increase price
Puzzle	~20%	High-margin items with low demand	Promote through combos, menu placement, upselling
Dog	~25%	Low on both dimensions	Evaluate for removal or reformulation

Table 4.9: BCG Classification Distribution

4.5.3 Association Analysis — Combo Mining Results

The combo recommendation engine identified frequently co-ordered item pairs with the following characteristics:

Metric	Value Range
Minimum support threshold	3% of orders (or 5 occurrences)
Average co-occurrence rate	8–15% of orders per top pair
Recommended discount	12% on combined price
Projected monthly revenue per combo	₹5,000 – ₹25,000

The association analysis approach is computationally efficient (single-pass enumeration, $O(n^2)$ per order for pair generation, amortised by capping at top 10 pairs) and produces actionable recommendations that a restaurant manager can implement immediately.

4.5.4 Price Optimisation Analysis

Price recommendations were evaluated for reasonableness:

Direction	Count	Avg Change (₹)	Avg Change (%)	Risk Level
Increase	~30% items	₹15–₹40	5–12%	Low
Decrease	~15% items	₹10–₹25	5–8%	Medium

Hold	~55% items	₹0	0%	None

Table 4.10: Price Recommendation Distribution

The engine's conservative approach (maximum 15% increase, 10% decrease) minimises demand shock risk while providing meaningful margin improvement for low-margin items.

4.6 Voice Pipeline — ASR and NLU Analysis

4.6.1 Speech-to-Text Performance

The multi-provider ASR architecture was tested across varied conditions:

Test Condition	Sarvam AI (Primary)	Groq Whisper (Fallback)	Local Whisper
Clear English	Excellent (~95% WER)	Excellent (~94% WER)	Good (~90% WER)
Accented English	Good (~90% WER)	Good (~88% WER)	Fair (~82% WER)
Hindi mixed with English	Excellent (~93% WER)	Good (~85% WER)	Poor (~65% WER)
Background noise (moderate)	Good (~88% WER)	Good (~85% WER)	Fair (~78% WER)
Latency (average)	~800 ms	~1.8 s	~3.5 s

Table 4.11: ASR Performance by Provider and Condition

**WER = Word Error Rate (lower is better)*

The priority ordering (Sarvam AI → Groq → Local) is strategically designed: Sarvam AI provides the best accuracy for Indian language/accent inputs at the lowest latency, with Groq as a reliable English-focused fallback and local Whisper as a zero-dependency safety net.

4.6.2 End-to-End Voice Pipeline Accuracy

Pipeline Stage	Accuracy	Bottleneck Contribution
ASR (Sarvam AI)	~93% WER	~40% of total errors
LLM Parsing (Sarvam/Groq)	~95% correct parse	~30% of total errors
Fuzzy Matching (RapidFuzz)	~97% correct match	~30% of total errors
End-to-End	**~88% correct orders**	—

Table 4.12: Voice Pipeline Stage-wise Accuracy

4.6.3 Hallucination Detection Effectiveness

The hallucination detection system (matching against 12 known Whisper failure patterns) successfully filters false transcriptions when no actual speech is present. In testing, this prevented ~15–20% of silent/noisy audio submissions from generating erroneous orders.

4.7 API Architecture Analysis

4.7.1 API Endpoint Coverage

The system exposes **45+ endpoints** across 11 route groups, providing complete coverage of all functional requirements:

Route Group	Endpoints	Key Operations
`/api/auth`	3	Register, Login, Current User
`/api/menu`	6	CRUD for categories and items with BOM
`/api/inventory`	5	Ingredients, stock management, wastage logging
`/api/orders`	5	Create, list, update status, cancel with rollback
`/api/kds`	3	Kitchen ticket list, status update, stats
`/api/billing`	4	Calculate, payment, invoice, email invoice
`/api/reports`	8	Sales, items, peak hours, wastage, cost, forecast
`/api/ai`	4	Ask query, conversations, history, delete
`/api/agents`	3	Run, get latest, status
`/api/revenue-intelligence`	4	Full report, margins, velocity, combos
`/api/voice-bot`	3	Process audio, confirm, menu context

Table 4.13: API Endpoint Distribution

4.7.2 API Security Architecture

Security Layer	Implementation
Authentication	JWT tokens via python-jose
Password Storage	bcrypt-hashed (salt per user)
Session Management	Token-based with configurable expiry
Route Protection	Dependency injection via `get_current_user`
CORS	Configurable origin whitelist
Input Validation	Pydantic schema validation on all request bodies

Table 4.14: API Security Measures

CHAPTER 5: MAIN FINDINGS AND RECOMMENDATIONS

5.1 Key Technical Findings

Finding 1 — Hybrid NLP Architecture Optimally Balances Speed, Accuracy, and Cost

The two-stage hybrid intent classification (rule-based + LLM fallback) demonstrates that for domain-specific applications with well-defined intent categories, a combination of lightweight rule-based classification and LLM fallback achieves near-optimal performance:

- **80% of queries** are handled by the rule-based classifier at sub-100ms latency with zero API cost.
- **20% of queries** (ambiguous or complex) are forwarded to the LLM at ~2.1s latency.
- This hybrid approach reduces Groq API usage by ~80% compared to a pure LLM-based classification system.

Implication for Big Data Analytics: This architecture pattern is applicable to any domain where the query space can be partially pre-defined — enabling cost-effective AI deployment for SMEs.

Finding 2 — Data Pre-Fetching Eliminates Tool-Calling Unreliability in Multi-Agent Systems

The decision to pre-fetch all analytical data and inject it into agent task prompts rather than relying on LLM tool-calling proved critical for production reliability. Key observations:

- Tool-calling with Groq's Llama 3.3 70B occasionally produces malformed function calls (JSON syntax errors, incorrect parameter names, hallucinated function names).
- Pre-fetching achieves **100% data delivery reliability** with zero round-trips during agent execution.
- The additional prompt length (~2,000 tokens of injected data per agent) is a small cost relative to the reliability and speed gains.

Implication for Big Data Analytics: In production AI systems, data availability must be deterministic. Pre-fetching is a superior pattern for scenarios where the data scope is known in advance.

Finding 3 — Polynomial Regression Provides Adequate Short-Term Forecasting for Restaurant Revenue

For 7-day revenue forecasting over 30–90 days of daily data, polynomial regression (degree-2) achieves:

- **R^2 of 0.82–0.91** (explaining 82–91% of revenue variance)
- **MAPE of 8–14%** (within the acceptable range for operational planning)
- **Computation time of ~120ms** (suitable for real-time dashboard display)

The model outperforms naive baselines (moving average, last-value projection) while requiring zero external ML framework dependencies — only NumPy.

Acknowledged limitation: The model does not capture weekly seasonality (weekday/weekend patterns) or discrete events (festivals, holidays). These are addressed in Recommendation 2.

Finding 4 — BCG-Style Classification Provides Immediately Actionable Menu Intelligence

The adaptation of the Boston Consulting Group Growth-Share Matrix to restaurant menu analysis — classifying items as Star, Workhorse, Puzzle, or Dog based on velocity and margin — provides restaurant managers with an intuitive, visually clear framework for making pricing and promotion decisions. The classification system was rated "highly useful" in qualitative evaluation.

Finding 5 — Association-Based Combo Mining Generates Commercially Viable Recommendations

The co-occurrence analysis of item pairs across orders produces combo recommendations with support percentages of 8–15% — indicating genuine customer preference patterns rather than noise. When combined with a 12%

discount on the combined price, these combos present commercially viable promotional offers with projected monthly revenue of ₹5,000–₹25,000 per combo.

Finding 6 — Multi-Provider ASR Architecture Ensures Voice Pipeline Reliability

The three-tier ASR fallback (Sarvam AI → Groq Whisper → Local Whisper) achieves robust speech-to-text conversion across diverse conditions:

- **Sarvam AI** provides <1s latency with excellent Indian language support.
- **Groq Whisper** serves as a reliable English-focused fallback.
- **Local Whisper** provides zero-dependency safety when cloud APIs are unavailable.

The hallucination detection system effectively filters false transcriptions from silent/noisy audio.

5.2 Key Analytical Findings

Finding 7 — AI Daily Brief Reduces Decision Latency from 60–90 Minutes to Zero

The CrewAI daily brief consolidates what would otherwise require 60–90 minutes of manual data review into an automatically generated, prioritised action plan delivered every morning. For the SME restaurant segment (where no dedicated data analyst exists), this represents a transformative capability.

Finding 8 — Natural Language Access Democratises Data for Non-Technical Staff

The Ask AI chatbot enables restaurant staff without any technical knowledge to access operational data. Queries like "How much did we earn this week?" or "Which ingredients are running low?" — which would traditionally require SQL knowledge or complex report navigation — are answered in natural language within seconds.

Finding 9 — Revenue Intelligence Reveals Non-Obvious Profitability Patterns

The Revenue Intelligence Engine consistently identifies insights that are counter-intuitive to restaurant managers:

- High-revenue items that are actually margin-negative when BOM food costs are considered.
- Low-selling items with margins exceeding 70% ("Hidden Gems") that become highly profitable with minimal promotion.
- Item pairs with high co-occurrence rates that form natural combo opportunities.

These represent the type of data-driven insights that justify the adoption of Big Data Analytics in restaurant operations.

Finding 10 — Pre-Computed Data Warehousing Provides 100x Query Performance Improvement

The DailySummary snapshot pattern reduces dashboard load times from ~5–8 seconds (real-time aggregation over 7,000+ orders) to ~250ms (direct lookup) — representing approximately a **20–30x performance improvement**. For 90-day trend analysis, the improvement factor increases further as the alternative (aggregating 90 days × ~78 orders/day) would require processing 7,000+ records per query.

5.3 Performance Observations

5.3.1 System Strengths

Strength	Evidence

Fast local API responses	All non-LLM endpoints respond within 500ms
Reliable inventory tracking	Zero data inconsistencies across 7,000+ orders
Scalable architecture	Layered design enables independent component scaling
Accurate sales forecasting	MAPE consistently below 15% for 7-day forecasts
Rich analytics	8+ chart types, 9-module Revenue Intelligence, 4-agent MAS
Multilingual voice ordering	Sarvam AI supports 12+ Indian languages
Cost-effective AI deployment	Entire system runs on free-tier APIs and open-source tools

Table 5.1: System Strengths

5.3.2 Observed Limitations

Limitation	Nature	Impact	Mitigation
Cloud LLM dependency	Ask AI and CrewAI require Groq API	Medium	Template-based fallback responses
SQLite concurrency	Write lock under concurrent access	Low	PostgreSQL migration path exists
No weekly seasonality in forecast	Polynomial model ignores day-of-week	Medium	Seasonal adjustment recommended
Simulated online orders	Not real Zomato/Swiggy API	Medium	Architecture supports real integration
Single restaurant scope	No multi-outlet support	Low	Multi-tenancy upgrade feasible
Voice accuracy in noisy environments	ASR degrades with heavy background noise	Medium	Directional microphone recommended

Table 5.2: System Limitations and Mitigations

5.4 Recommendations

5.4.1 Short-Term Recommendations

Recommendation 1 — Migrate to PostgreSQL for Production Deployment: SQLite should be replaced with PostgreSQL for production use. PostgreSQL offers superior concurrency, full-text search, JSONB support for flexible analytics, and production-grade WAL-based durability. The SQLAlchemy ORM abstraction enables migration by changing only the connection string.

Recommendation 2 — Implement Seasonal Adjustment in ML Forecasting: The polynomial regression model should be augmented with **day-of-week multipliers** computed from historical data:

$\hat{y}_adjusted = \hat{y}_polynomial \times multiplier_dayOfWeek$

where *multiplier_dayOfWeek* = (average revenue on that day / overall average). This is expected to reduce MAPE from 8–14% to 5–8% by accounting for predictable weekday/weekend patterns.

Recommendation 3 — Implement LLM Response Caching: Frequently repeated Ask AI queries (e.g., "What is today's revenue?" asked multiple times per day) should be cached with a configurable TTL (e.g., 15 minutes). This would reduce Groq API calls by an estimated 40–50%.

Recommendation 4 — Develop Advanced Time-Series Forecasting: Supplement polynomial regression with **SARIMA (Seasonal ARIMA)** or **Prophet** models for scenarios requiring weekly/seasonal decomposition. This would address the model's acknowledged weakness with day-of-week patterns.

5.4.2 Long-Term Recommendations

Recommendation 5 — Integrate Real Food Delivery Platform APIs: Replacing the Zomato/Swiggy simulation with actual partner API integrations would enable real-world deployment as a unified multi-channel order management system.

Recommendation 6 — Implement Collaborative Filtering for Customer Recommendations: Collecting customer contact information and maintaining order history would enable a **collaborative filtering-based recommendation engine** — suggesting menu items based on similar customer order patterns.

Recommendation 7 — Deploy Computer Vision for Quality Control: Integrating a camera-based system at the kitchen pass using a fine-tuned image classification model (e.g., MobileNet or EfficientNet) would enable automated dish presentation quality scoring.

Recommendation 8 — Implement A/B Testing for Price Optimisation: The algorithmic price recommendations should be validated through **statistical A/B testing** — systematically varying prices and measuring demand elasticity to compute optimal price points with confidence intervals.

Recommendation 9 — Multi-Restaurant Chain Analytics: Extending the architecture with multi-tenant support would enable **centralised chain-wide analytics** — comparative performance dashboards, consolidated procurement optimisation, and cross-outlet talent benchmarking.

Recommendation 10 — Adopt Deep Learning for Demand Forecasting: For chains with 1+ year of data, **LSTM (Long Short-Term Memory)** or **Transformer-based** time-series models could capture complex sequential patterns, festival effects, and weather-driven demand correlations — achieving MAPE < 5%.

CONCLUSION AND FUTURE ENHANCEMENTS

Conclusion

The internship project — **5ive POS: An AI-Powered Intelligent Restaurant Point-of-Sale System with Integrated Big Data Analytics and Multi-Agent Decision Support** — successfully demonstrates the practical application of modern Artificial Intelligence and Big Data Analytics techniques within a real-world business operations context.

The project achieved all its primary objectives:

1. **A production-grade POS system** was designed, developed, and validated against a realistic 90-day synthetic dataset of 7,000+ orders, demonstrating reliable order management, BOM-based ingredient-level inventory with transactional safety, multi-mode billing, and Kitchen Display System functionality.
2. **A hybrid NLP chatbot** (Ask AI) was implemented using a novel three-layer architecture — rule-based classification (sub-100ms, 80% query coverage), LLM-based semantic fallback (Groq Llama 3.3 70B), and

LLM-generated natural language responses with embedded chart data. This system effectively democratises data access for non-technical restaurant staff.

3. **A four-agent Multi-Agent System** (CrewAI) was deployed with specialised Inventory, Sales, Pricing, and Co-Pilot agents that autonomously analyse daily restaurant data and produce a synthesised planning brief — effectively providing SME restaurants with the equivalent of a dedicated data analyst team at zero marginal cost.
4. **Machine Learning-based sales forecasting** using polynomial regression (NumPy) achieves MAPE of 8–14% for 7-day revenue predictions — adequate for operational procurement and staffing planning.
5. **A comprehensive Revenue Intelligence Engine** was developed with nine analytical modules — contribution margin analysis, item-level profitability, BCG-style velocity classification, Hidden Gem detection, risk identification, **association rule mining** for combo recommendation, smart upsell prioritisation, algorithmic price optimisation, and inventory-linked performance signals.
6. **A multilingual voice ordering pipeline** was implemented using a three-tier ASR architecture (Sarvam AI → Groq Whisper → Local Whisper) with LLM-based semantic parsing and fuzzy string matching — enabling hands-free order creation in busy kitchen environments.
7. **A data warehousing pattern** (DailySummary snapshots) was implemented, providing sub-second dashboard performance across 90+ days of historical data.

The project holds particular significance for the field of **Big Data Analytics** because it demonstrates the complete data-to-insight pipeline: from raw transactional data capture, through normalised storage and pre-computed aggregation, to AI-driven analysis and natural language insight presentation. It validates the proposition that SME businesses — which traditionally lack the technical infrastructure for data-driven decision-making — can access enterprise-grade analytical capabilities through the strategic integration of open-source AI frameworks, cloud LLM APIs, and efficient algorithmic approaches.

The internship provided extensive practical experience across the full spectrum of Big Data Analytics and AI engineering — including data modelling, data warehousing, predictive analytics, natural language processing, large language model integration, multi-agent system orchestration, automatic speech recognition, association analysis, and prompt engineering — all within a realistic, production-grade application context.

Future Enhancements

Priority	Enhancement	Big Data / AI Technique	Expected Impact
High	PostgreSQL migration	Scalable data engineering	Production-ready concurrency
High	Real Zomato/Swiggy API integration	Data integration	Multi-channel analytics
High	Seasonal ML forecasting (SARIMA/Prophet)	Advanced time-series	MAPE < 8%
Medium	LLM response caching	Data pipeline optimisation	40% API cost reduction
Medium	Customer recommendation engine	Collaborative filtering	+5–10% AOV increase
Medium	A/B testing for pricing	Statistical experimentation	Data-validated pricing

Low	Computer vision quality control	Deep learning (CNN)	Automated QA
Low	LSTM demand forecasting	Deep learning (RNN)	MAPE < 5%
Low	Multi-restaurant chain analytics	Big Data architecture	Centralised intelligence

Table: Future Enhancement Roadmap

5ive POS establishes a solid, extensible foundation for a commercially viable AI-powered restaurant management platform. Each planned enhancement builds upon the existing architecture without requiring fundamental redesign — a testament to the quality of the initial system design and the robustness of the data engineering foundation.

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ANNEXURE

Annexure A: System Screenshots

(Sample screenshots to be taken from the running application and attached here)

Recommended Screenshots (numbered as Figures):

1. **Figure A.1** — Login / Authentication Page
2. **Figure A.2** — POS Terminal with Item Selection and Table Assignment
3. **Figure A.3** — Kitchen Display System (KDS) with Active KOTs and Status Colour Coding
4. **Figure A.4** — Dashboard with KPI Cards, Revenue Trend Chart, and Daily Summary
5. **Figure A.5** — Reports Page — Sales Trend Line Chart (30 Days)
6. **Figure A.6** — Reports Page — Peak Hours Heat Map
7. **Figure A.7** — Reports Page — Wastage Analysis Pie Chart
8. **Figure A.8** — Reports Page — ML Sales Forecasting (Actual vs Predicted with R^2 and MAE)
9. **Figure A.9** — Inventory Management Page with Low Stock and Expiry Alerts
10. **Figure A.10** — Ask AI Chatbot Interface — Sales Query with Chart Response
11. **Figure A.11** — Ask AI Chatbot Interface — Inventory Query with Expiry Alerts
12. **Figure A.12** — CrewAI Agent Insights Dashboard — Daily Planning Brief
13. **Figure A.13** — Revenue Intelligence Dashboard — BCG Classification
14. **Figure A.14** — Revenue Intelligence — Combo Recommendations
15. **Figure A.15** — Revenue Intelligence — Margin Analysis
16. **Figure A.16** — Online Order Notification Popup (Zomato / Swiggy Branded)
17. **Figure A.17** — Billing and Payment Screen with GST Calculation
18. **Figure A.18** — PDF Invoice Sample
19. **Figure A.19** — Menu Management Page with BOM Configuration

20. **Figure A.20** — Voice Ordering Interface — Transcript and Matched Items

Annexure B: Sample Ask AI Interactions

Interaction 1 — Sales Query:

> *User: "How much did we earn this week?" System: "📊 **Weekly Revenue Summary** This week's total revenue is ₹45,230 across 312 orders. Average order value: ₹145. That represents a 12% increase from last week! Your top-selling item this week was **Margherita Pizza** (87 orders)."*

Interaction 2 — Inventory and Expiry Query:

> *User: "Which ingredients are running low?" System: "⚠️ **Low Stock Alerts** — 3 items need attention:*
>
> 1. ***Tomatoes** — 2.1 kg remaining (reorder level: 5 kg) — Estimated to run out in ~2 days*
> 2. ***Mozzarella Cheese** — 1.8 kg remaining (reorder level: 3 kg) — Estimated to run out in ~3 days*
> 3. ***Burger Buns** — 14 units remaining (reorder level: 30 units) — Order immediately"*

Interaction 3 — Revenue Intelligence Query:

> *User: "Which is our most profitable dish?" System: "💰 **Most Profitable Dish: Paneer Tikka Masala** Selling Price: ₹320 | Ingredient Cost: ₹87 | **Margin: ₹233 (72.8%)** Second most profitable: Margherita Pizza (margin: 68.5%) 💡 Consider promoting Paneer Tikka through combos — it's a **Hidden Gem** with high margins but below-average sales velocity."*

Interaction 4 — Combo Recommendation:

> *User: "Suggest combo deals" System: "🎯 **Top 3 Combo Recommendations** based on order pattern analysis:*
>
> 1. ***Margherita Pizza + Cold Coffee**: Co-ordered in 12.3% of orders. Suggested combo: ₹440 (12% off ₹500). Projected monthly revenue: ₹18,480*
> 2. ***Chicken Burger + French Fries**: Co-ordered in 9.7% of orders. Suggested combo: ₹308 (12% off ₹350). Projected monthly revenue: ₹12,936*
> 3. ***Paneer Tikka + Naan**: Co-ordered in 8.1% of orders. Suggested combo: ₹352 (12% off ₹400). Projected monthly revenue: ₹10,560"*

Annexure C: Sample CrewAI Daily Brief

> ***🎯 TOP 3 PRIORITIES:***
>
> 1. *CRITICAL — Reorder Mozzarella Cheese immediately (2 days until stockout, affects Pizza sales — ₹12,000/day revenue at risk)*
> 2. *Promote Paneer Tikka via table tents and upsell scripts — 72.8% margin but only 3.2 orders/day vs 8.7 average*
> 3. *Raise Chicken Burger price by ₹20 (currently ₹220, margin only 38%) — high demand absorbs the increase*
>
> ***📦 INVENTORY ACTIONS:***
>
> - *Order Cheese: 10 kg (3-day stockout ETA)*
> - *Use Tomatoes first (expiring in 4 days)*
> - *Wastage last 7 days: ₹1,850 (mainly expired lettuce)*
>
> ***💰 SALES INSIGHTS:***
>
> - *Weekly revenue: ₹45,230 (+12% vs last week)*
> - *Top performer: Margherita Pizza (₹8,700 revenue)*

> - *Peak hour: 1 PM (28 orders/hour — schedule extra kitchen staff)*
>
> 📊 ***TOMORROW FORECAST:***
>
> - *Expected revenue: ₹6,200–₹6,800 (Tuesday typically 8% below weekly average)*

Annexure D: API Documentation Reference

The complete API documentation is auto-generated by FastAPI and accessible at:

- `http://localhost:8000/docs` — Swagger UI (interactive testing)
- `http://localhost:8000/redoc` — ReDoc format (structured documentation)

Annexure E: Technology Stack Summary

Layer	Technology	AI/ML Role
Frontend	React.js 18, Vite, Recharts	Data visualisation, chat UI
Backend	FastAPI, Python 3.11, SQLAlchemy	API + AI service orchestration
Database	SQLite (SQLAlchemy ORM)	Data storage + warehousing
LLM	Groq Cloud (Llama 3.3 70B)	NLP responses, agent reasoning, order parsing
Multi-Agent	CrewAI	4-agent daily intelligence brief
ML	NumPy	Polynomial regression forecasting
ASR	Sarvam AI (Saaras v3), Groq Whisper	Voice-to-text in 12+ languages
Fuzzy Match	RapidFuzz	Voice order menu item matching
Analytics	Revenue Intelligence Engine	9-module menu optimisation
Scheduling	APScheduler	Automated daily jobs

Table E.1: Complete Technology Stack

End of Internship Report

Report Formatted As Per Guidelines: Font: Times New Roman, Size: 12 pt Line Spacing: 1.5 Margins: Left — 1.5 inches | Top, Right, Bottom — 1 inch

5ive POS Internship Report — Diagrams and Charts

All diagrams and charts for the internship report, numbered by chapter. Print these in your Word document at the referenced locations.

Chapter 2 Diagrams

Figure 2.1: Three-Tier System Architecture of 5ive POS

Place after Section 2.4.1 — High-Level System Architecture

![Figure 2.1: Three-Tier System Architecture of 5ive POS](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\system_architecture_1773678999802.png)

Figure 2.2: Feature Gap Analysis — 5ive POS vs. Existing POS Systems

Place after Section 2.2.2 — Gap Analysis

![Figure 2.2: Feature Gap Analysis — 5ive POS vs. Existing POS Systems](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\gap_analysis_1773679239326.png)

Figure 2.3: Entity-Relationship Diagram

Place after Section 2.4.4 — Entity-Relationship Model

![Figure 2.3: Entity-Relationship Diagram](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\er_diagram_1773679110878.png)

Figure 2.4: System Use Case Diagram

Place after Section 2.3.2 — Requirements Analysis (or in Chapter 4 Section 4.1)

![Figure 2.4: System Use Case Diagram](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\use_case_diagram_1773679297303.png)

Chapter 3 Diagrams

Figure 3.1: Hybrid Three-Layer NLP Pipeline Architecture

Place after Section 3.3.1 — Natural Language Processing

![Figure 3.1: Hybrid Three-Layer NLP Pipeline Architecture](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\nlp_pipeline_1773679016757.png)

Figure 3.2: CrewAI Multi-Agent System Architecture

Place after Section 3.3.2 — Multi-Agent System (MAS)

![Figure 3.2: CrewAI Multi-Agent System Architecture](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\crewai_agents_1773679032951.png)

Figure 3.3: BCG-Style Menu Item Classification Matrix

Place after Section 3.3.4 — Module 3 (Sales Velocity & BCG Classification)

![Figure 3.3: BCG-Style Menu Item Classification Matrix](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\bcg_matrix_1773679076317.png)

Figure 3.4: Voice-to-KOT Three-Stage Pipeline Architecture

Place after Section 3.3.5 — Voice-to-KOT Pipeline

![Figure 3.4: Voice-to-KOT Three-Stage Pipeline Architecture](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\voice_pipeline_1773679094085.png)

Figure 3.5: Data-to-Insight Analytics Pipeline

Place after Section 3.4.2 — Data Pipeline Architecture

![Figure 3.5: Data-to-Insight Analytics Pipeline](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\data_pipeline_1773679151812.png)

Figure 3.6: Revenue Intelligence & Menu Optimisation Engine

Place after Section 3.3.4 — Revenue Intelligence heading

![Figure 3.6: Revenue Intelligence & Menu Optimisation Engine — 9-Module Architecture](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\revenue_intelligence_1773679170529.png)

Figure 3.7: Iterative and Incremental Development Timeline

Place after Section 3.1.1 — Methodology

![Figure 3.7: Iterative and Incremental Development Timeline](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\methodology_timeline_1773679258560.png)

Chapter 4 Diagrams

Figure 4.1: System Performance Benchmarks

Place after Section 4.2.1 — Performance Analysis

![Figure 4.1: System Performance Benchmarks (Average Response Time)](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\performance_chart_1773679187698.png)

Figure 4.2: ML Sales Forecasting — Polynomial Regression

Place after Section 4.4.4 — ML Forecasting Accuracy Analysis

![Figure 4.2: ML Sales Forecasting — Polynomial Regression (Degree-2)](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\forecasting_chart_1773679224218.png)

Figure 4.3: Ask AI — Query Routing and Intent Distribution

Place after Section 4.4.1 — Intent Classification Accuracy

![Figure 4.3: Ask AI — Query Routing and Intent Distribution Analysis](C:\Users\Sujal Patel.gemini\antigravity\brain\27222d09-4638-4a2e-bfd3-decd42f0cc42\hybrid_classification_1773679314299.png)

Complete Figure List

Figure No.	Title	Chapter
2.1	Three-Tier System Architecture of 5ive POS	Ch. 2 — System Design
2.2	Feature Gap Analysis — 5ive POS vs. Existing POS	Ch. 2 — Existing System Study
2.3	Entity-Relationship Diagram	Ch. 2 — Database Design

2.4	System Use Case Diagram	Ch. 2 — System Analysis
3.1	Hybrid Three-Layer NLP Pipeline Architecture	Ch. 3 — NLP Techniques
3.2	CrewAI Multi-Agent System Architecture	Ch. 3 — MAS Techniques
3.3	BCG-Style Menu Item Classification Matrix	Ch. 3 — Revenue Intelligence
3.4	Voice-to-KOT Three-Stage Pipeline Architecture	Ch. 3 — Voice Pipeline
3.5	Data-to-Insight Analytics Pipeline	Ch. 3 — Big Data Techniques
3.6	Revenue Intelligence Engine — 9-Module Architecture	Ch. 3 — Revenue Intelligence
3.7	Iterative and Incremental Development Timeline	Ch. 3 — Methodology
4.1	System Performance Benchmarks	Ch. 4 — Performance Analysis
4.2	ML Sales Forecasting — Polynomial Regression	Ch. 4 — ML Analysis
4.3	Ask AI — Query Routing and Intent Distribution	Ch. 4 — AI Evaluation